
Supplementary Data for Layered representation diagnostics for ultra-short metagenomic reads

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Abstract

Supplementary figures and source tables supporting the representation-diagnostic analyses, including baseline, estimator, perturbation, parameter, redundancy and runtime audits.

Keywords supplementary data, metagenomics, representation diagnostics

Supplementary data overview

This supplementary file collects the visual audits and source-table preview that support the main representation-diagnostic manuscript. Figures S1–S3 evaluate baseline controls, empirical information proxies and sequencing-error-aware perturbations. Figures S4–S7 examine local mutation fraction, P-channel counterfactuals, MSP bin and weight sensitivity, kNN mutual-information robustness and dimension-matched high-k compressed baselines. Figures S8–S10 summarize P/MSP redundancy, runtime, relation audits and the CAMI II marine anonymous-read stability probe. Table S1 gives a compact preview of the CAMI II marine source table; Table S2 reports all prespecified conditional-contribution contrasts from the seven-group K/P/MSP ablation.

Supplementary Figures

Supplementary Tables

Supplementary Figure S1 | Baseline and mixed-metric audit

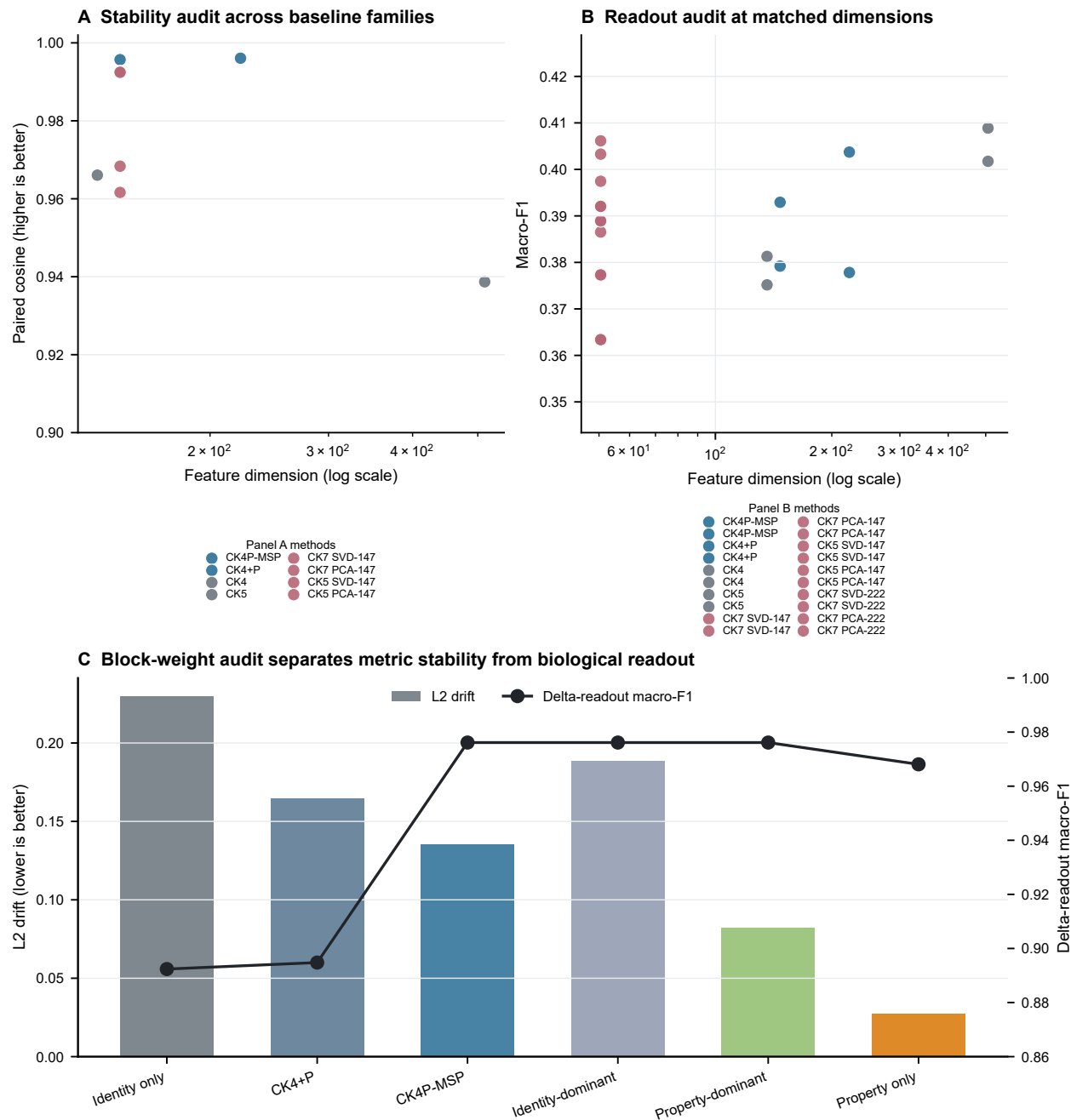


Figure S1 Baseline and mixed-metric audit. Panels compare same-dimension PCA/SVD controls, mixed-representation stability and block-weight settings on matched perturbation cells. These controls test whether the main drift pattern can be attributed to feature count, dimensional reduction or one selected K/P/MSP weighting.

Alt text: Baseline audit plots comparing full-position matrices, reduced controls and block-weight sensitivity.

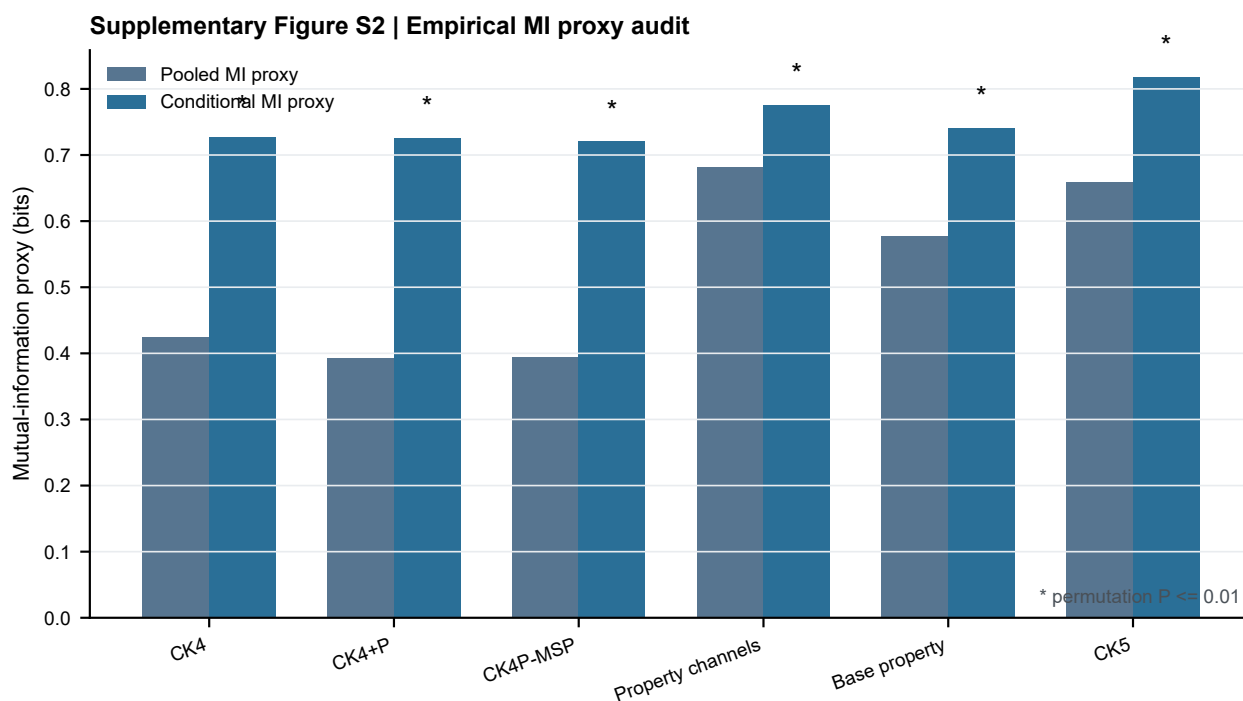


Figure S2 Estimator-dependent mutual-information (MI) and conditional-MI audit for local change versus matched nuisance perturbation. Values summarize low-dimensional representation-derived distance features; they are descriptive empirical estimates rather than universal information bounds.

Alt text: Bar plots summarizing empirical mutual-information and conditional-mutual-information audit values across representation blocks.

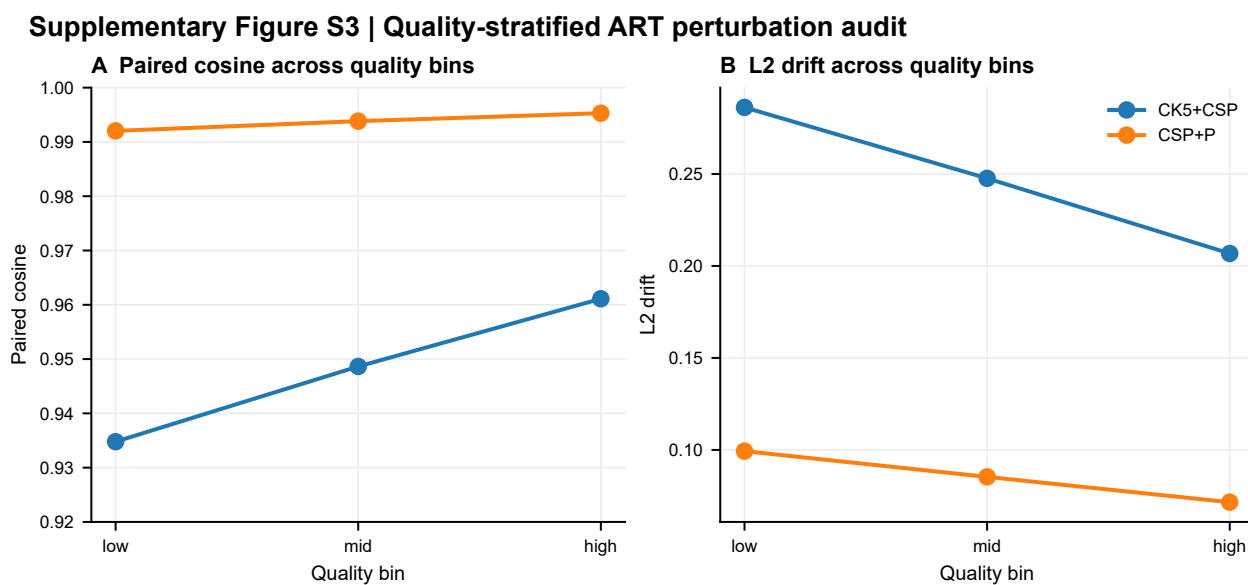


Figure S3 Quality-stratified ART perturbation audit. Paired cosine similarity and standardized L2 drift are summarized across low-, middle- and high-quality simulator-derived read strata to test whether the compact stability pattern depends on pooling all ART errors.

Alt text: Line plots showing ART perturbation behavior across quality levels for paired cosine and L2 drift.

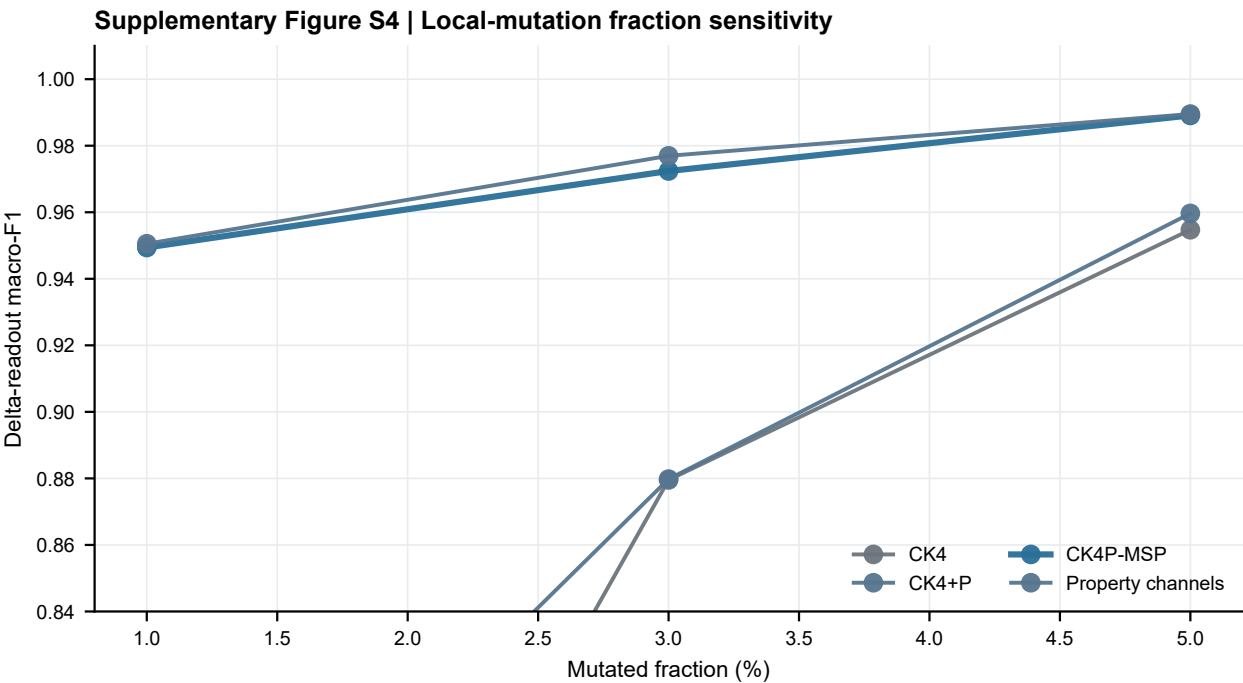


Figure S4 Local mutation-fraction sweep across compact and full-position representations. Curves report the response to increasing structured local-change fractions relative to matched nuisance perturbation; the sweep tests sensitivity across effect sizes rather than a functional-variant classification task.

Alt text: Line plot showing local mutation-fraction sensitivity across compact representation families.

Table S1. Preview of the CAMI II marine subset source table. The full CSV is retained in the data package.

length	condition	method	features	paired cosine	L2 drift	top-1
69	N_3pct	CK4	136	0.9643	0.2651	1.0000
69	N_3pct	CK4+P	147	0.9944	0.1059	1.0000
69	N_3pct	CK4P-MSP	222	0.9948	0.1016	1.0000
69	N_3pct	CK4P-MSP+SD	297	0.9943	0.1061	1.0000
69	N_3pct	CK5	512	0.9382	0.3491	1.0000
69	N_3pct	CK5+CK4P-MSP	734	0.9665	0.2572	1.0000
69	N_3pct	CSP	147	0.9941	0.1080	1.0000
75	N_3pct	CK4	136	0.9683	0.2497	1.0000
75	N_3pct	CK4+P	147	0.9950	0.0992	1.0000
75	N_3pct	CK4P-MSP	222	0.9954	0.0951	1.0000
75	N_3pct	CK4P-MSP+SD	297	0.9950	0.0994	1.0000
75	N_3pct	CK5	512	0.9441	0.3322	1.0000
75	N_3pct	CK5+CK4P-MSP	734	0.9698	0.2444	1.0000
75	N_3pct	CSP	147	0.9948	0.1011	1.0000
100	N_3pct	CK4	136	0.9687	0.2483	1.0000

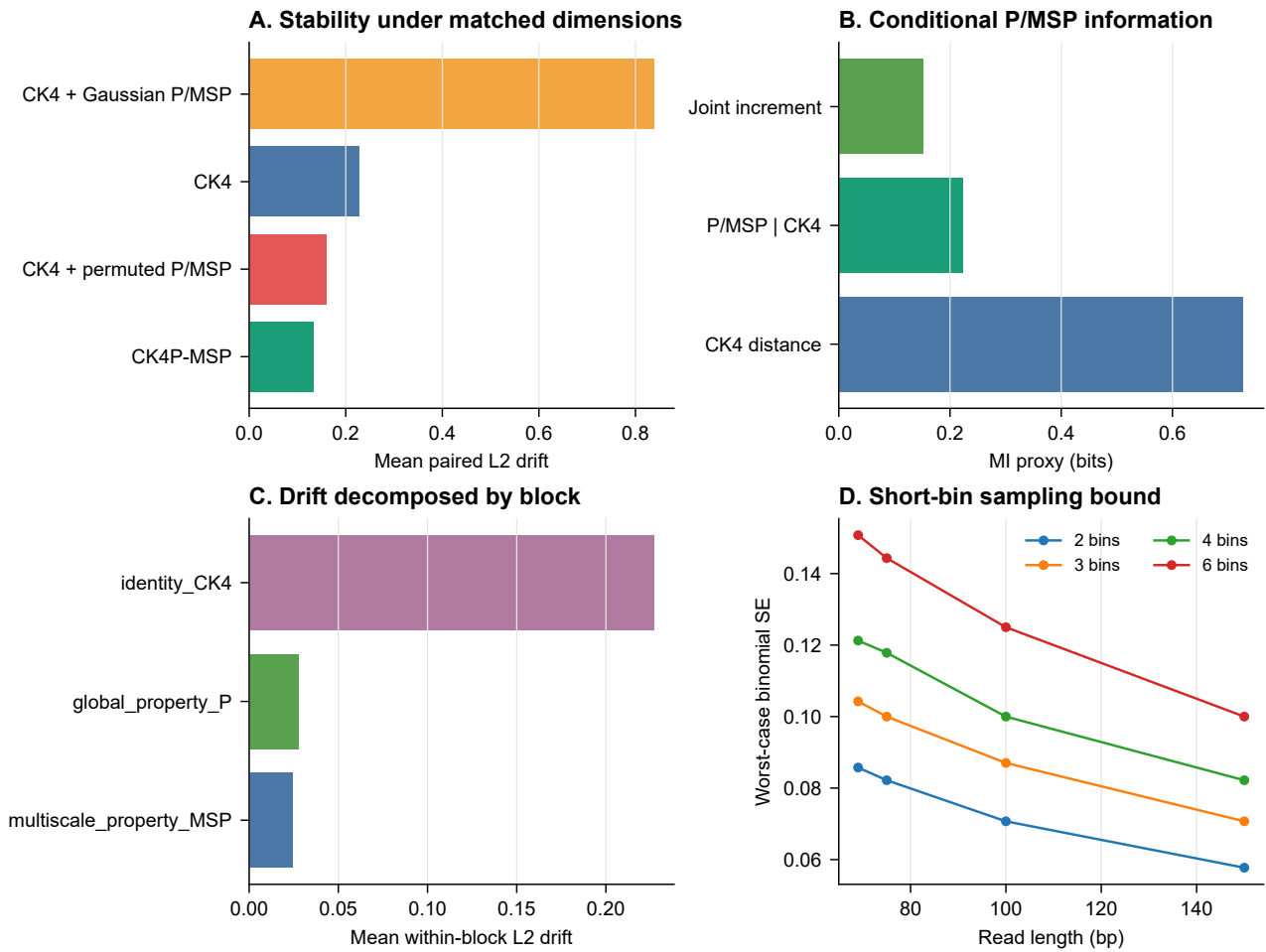


Figure S5 P-channel counterfactual and short-bin reliability audit. Sequence-linked P features are compared with permuted or random added channels, blockwise drift is decomposed, and finite-bin sampling bounds are shown for the shortest read conditions.

Alt text: Multi-panel audit of P-channel counterfactual behavior, drift decomposition and short-bin sampling bounds.

Table S2. Prespecified conditional-contribution contrasts from the seven-group K/P/MSP ablation. The full representation is CK4P-MSP. Improvement is signed so that positive values favour the full representation; for L2 drift, the sign is reversed because lower values are preferable. Confidence intervals are percentile intervals from 10,000 paired-cell bootstrap resamples. Two-sided Wilcoxon signed-rank tests were adjusted across all 12 rows by the Benjamini–Hochberg procedure. Global cells are length-by-perturbation pairs; local cells are length-by-local-mode grouped cross-validation summaries.

block contrast	evidence layer	metric	<i>n</i>	full	comparator	improvement [95% CI]; BH <i>q</i>
K P+MSP	global	paired cosine	24	0.989	0.999	−0.010 [−0.013, −0.008]; 2.4×10^{-7}
K P+MSP	global	L2 drift	24	0.129	0.027	−0.101 [−0.118, −0.086]; 2.4×10^{-7}
K P+MSP	global	retrieval	24	1.000	0.942	0.058 [0.036, 0.081]; 7.3×10^{-4}
K P+MSP	local	grouped macro-F1	12	0.979	0.970	0.008 [−0.0002, 0.019]; 0.507
P K+MSP	global	paired cosine	24	0.989	0.984	0.005 [0.004, 0.006]; 2.4×10^{-7}
P K+MSP	global	L2 drift	24	0.129	0.156	0.027 [0.023, 0.031]; 2.4×10^{-7}
P K+MSP	global	retrieval	24	1.000	1.000	0.000 [0.000, 0.000]; 1.000
P K+MSP	local	grouped macro-F1	12	0.979	0.978	0.001 [−0.0002, 0.002]; 0.525
MSP K+P	global	paired cosine	24	0.989	0.984	0.005 [0.004, 0.006]; 2.4×10^{-7}
MSP K+P	global	L2 drift	24	0.129	0.156	0.027 [0.023, 0.032]; 2.4×10^{-7}
MSP K+P	global	retrieval	24	1.000	1.000	0.000 [0.000, 0.000]; 1.000
MSP K+P	local	grouped macro-F1	12	0.979	0.904	0.074 [0.035, 0.117]; 7.3×10^{-4}

Supplementary Figure S6 | MSP binset and gamma-sensitivity audit

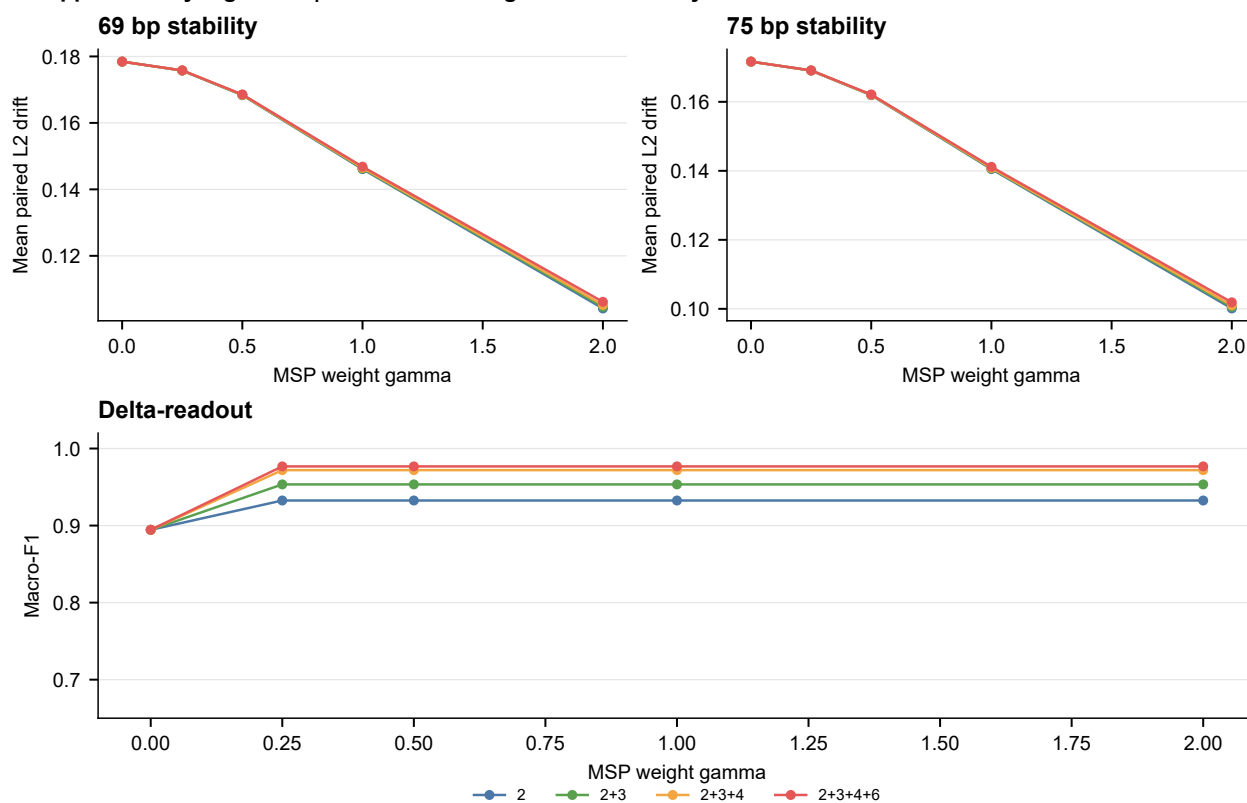


Figure S6 MSP binset and weight-sensitivity audit at 69 and 75 bp. Alternative coarse-to-fine binsets and $\gamma \in \{0, 0.25, 0.5, 1, 2\}$ are compared using stability and grouped local-change readout, testing whether the default MSP result depends on one fragile bin or weight setting.

Alt text: Line plots showing MSP binset and gamma-weight sensitivity for 69 and 75 bp reads and delta-readout.

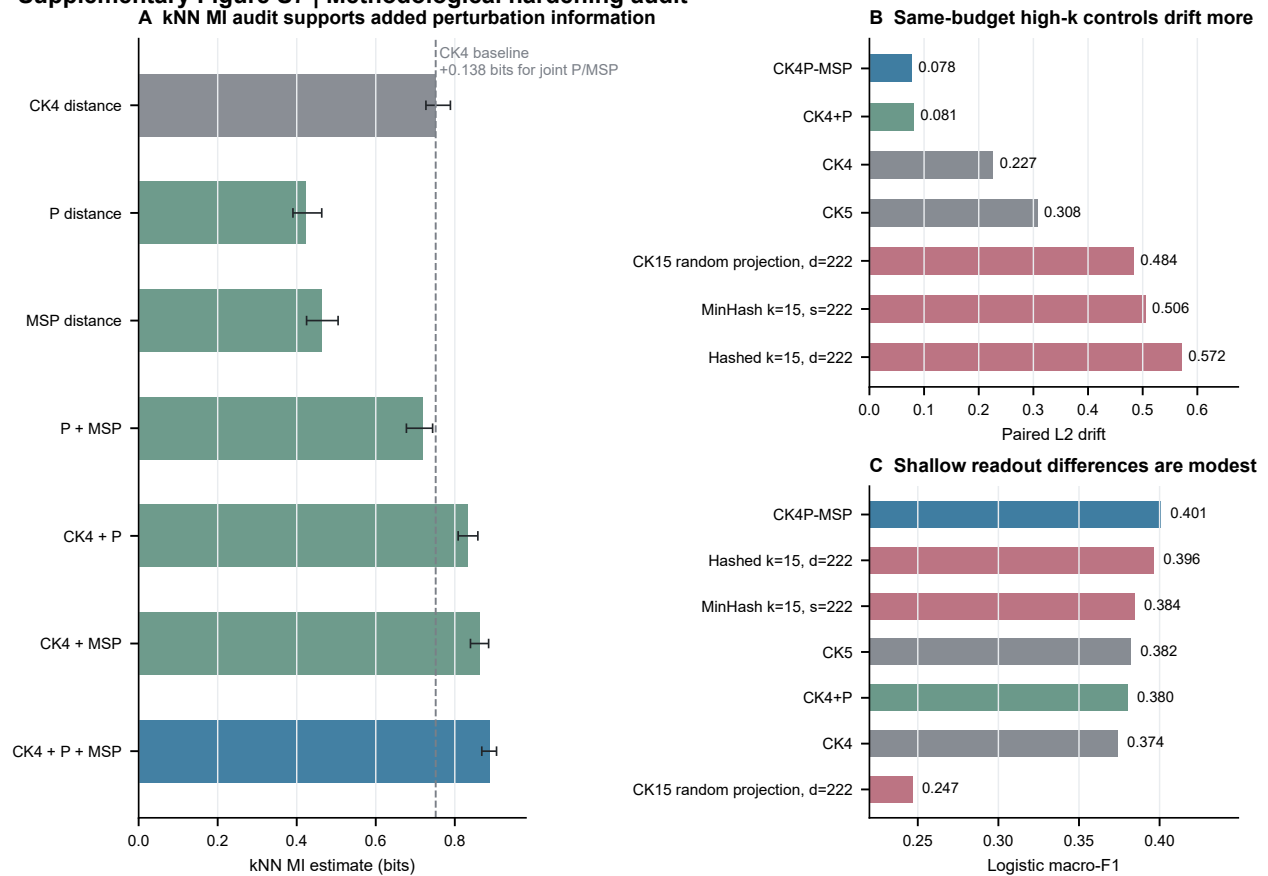
Supplementary Figure S7 | Methodological hardening audit

Figure S7 k-nearest-neighbour MI robustness and dimension-matched high-k compressed baseline audit. Estimator settings and subsamples test the stability of the empirical MI increment; hashed and sparse-random-projection canonical $k = 15$ vectors test perturbation stability at the 222-feature CK4P-MSP budget.

Alt text: Multi-panel audit showing kNN mutual-information robustness and dimension-matched high-k compressed baseline comparisons.

Supplementary Figure S8 | P/MSP contribution, redundancy and runtime audit

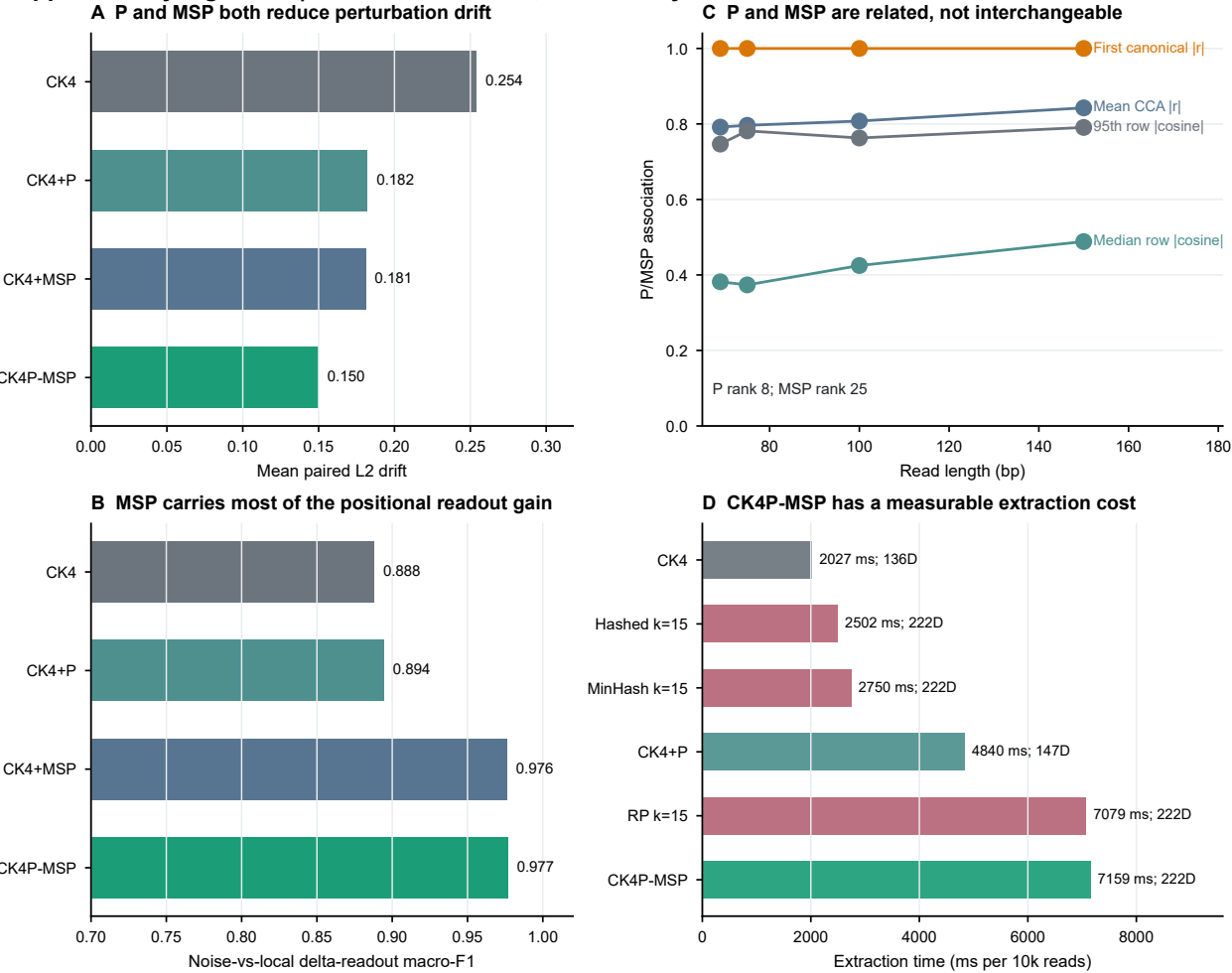


Figure S8 P/MSP contribution, redundancy and runtime audit. Blockwise contribution summaries are shown alongside relation measures and feature-extraction time, separating scientific contribution from computational cost. The relation results support overlapping but non-equivalent P and MSP channels.

Alt text: Multi-panel audit showing P and MSP contributions, redundancy and runtime costs.

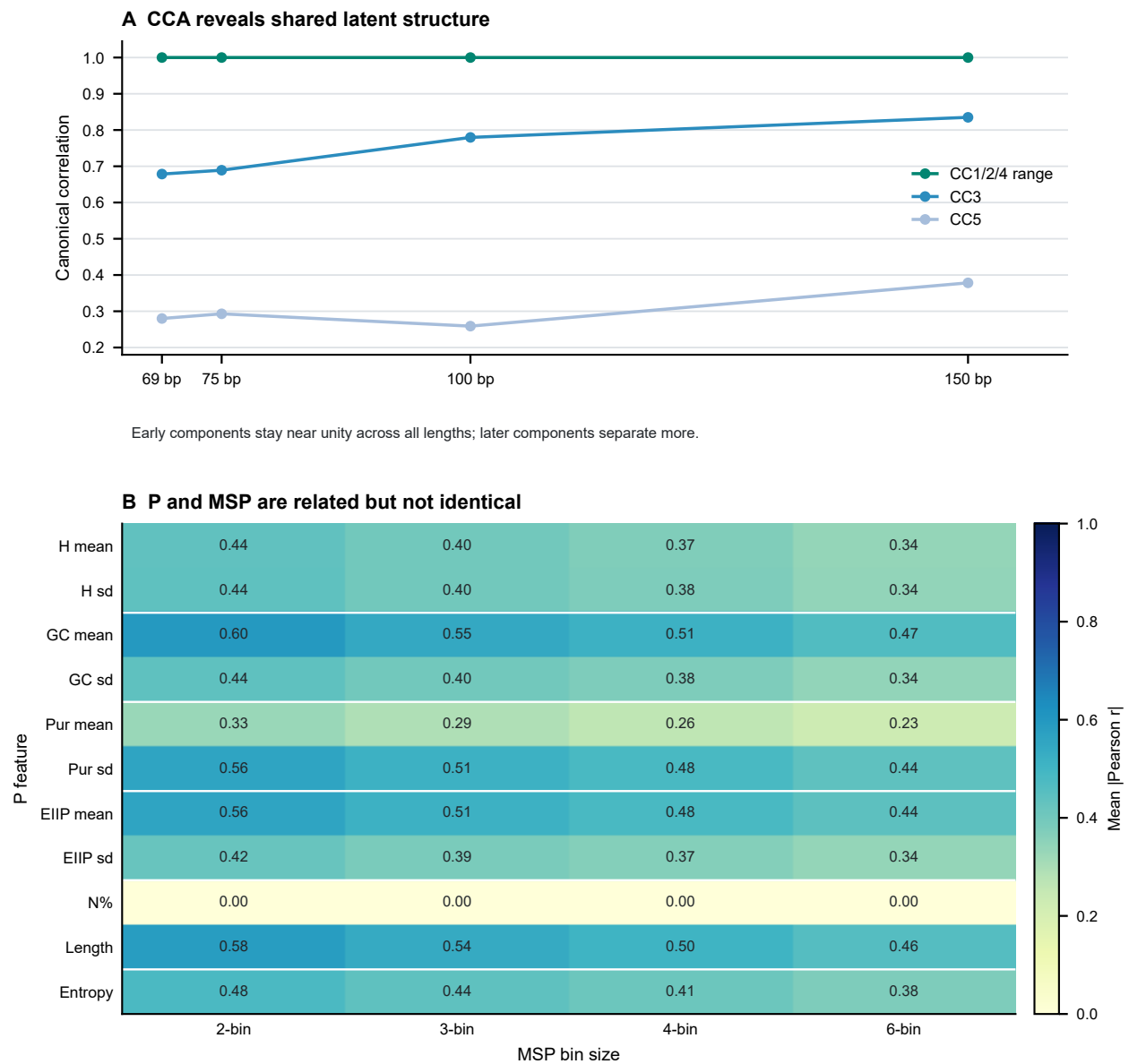


Figure S9 P/MSP relation audit across the tested read lengths. Canonical-correlation, rank and row-wise alignment summaries quantify the shared biochemical component and the additional multiscale variation in MSP; they are not tests of statistical independence.

Alt text: Line and heatmap panels showing CCA and correlation relationships between P and MSP feature layers.

Supplementary Figure S10 | CAMI II marine subset probe
n=4,000 reads; 48,000 length/condition rows; unlabeled perturbation-stability probe

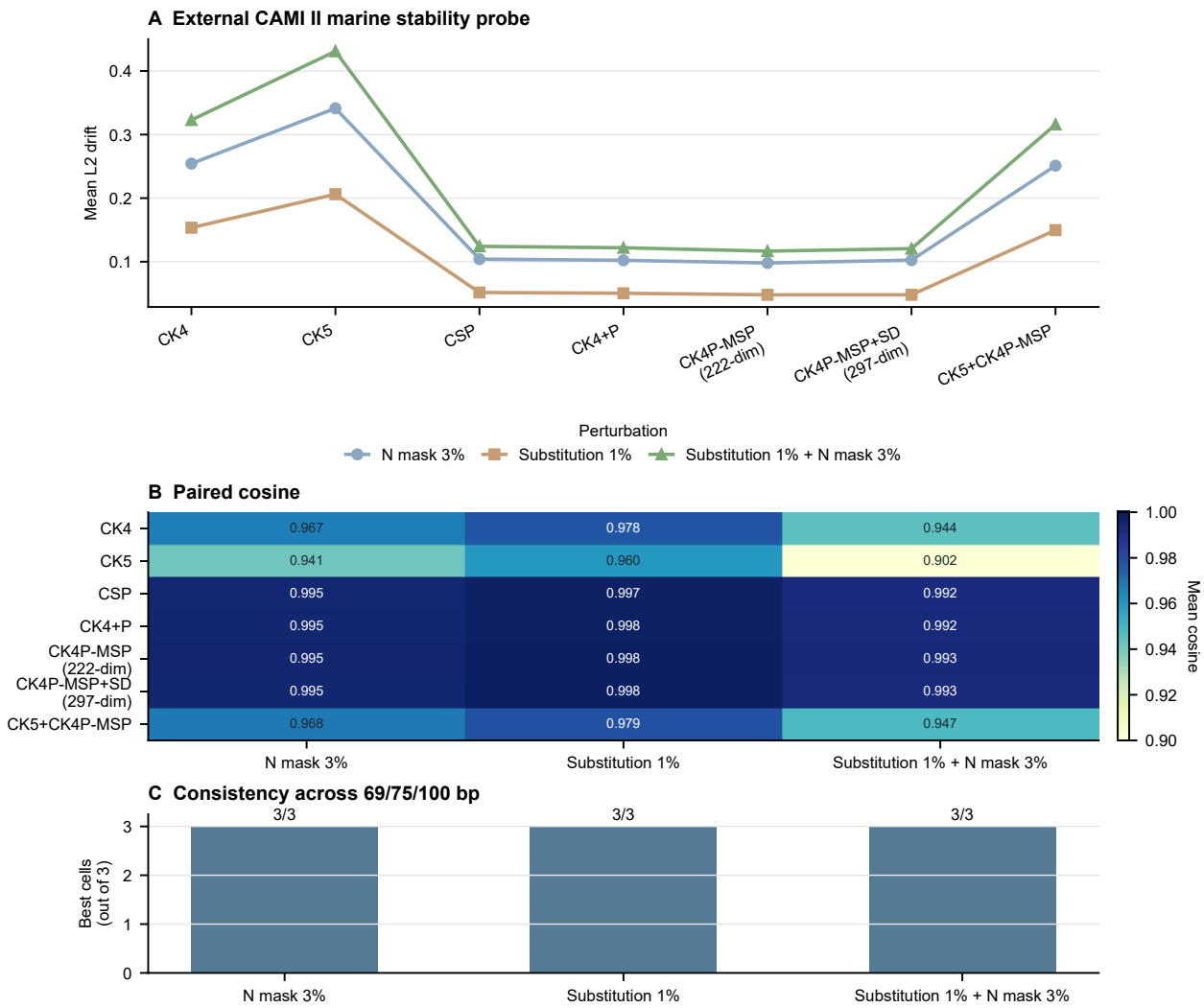


Figure S10 CAMI II marine anonymous-read stability probe. Paired cosine similarity, standardized L2 drift and nearest-clean retrieval are summarized for 4,000 anonymous source reads truncated to 69, 75 and 100 bp under N masking, substitution and combined perturbations. Read-level taxonomic labels were not reconstructed, so this figure supports external stability only.

Alt text: Line, heatmap and bar panels showing CAMI II marine anonymous-read stability across length and perturbation settings.